

Complex Numbers

$$z = \underset{\substack{\uparrow \\ \text{real}}}{a} + \underset{\substack{\uparrow \\ \text{imaginary}}}{bi}$$

Cartesian to Polar

$$(x, y) \rightarrow r, \theta$$

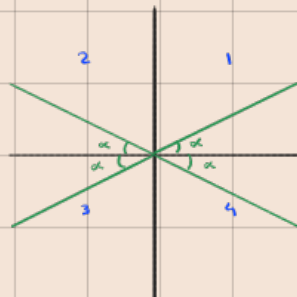
$$r = \sqrt{x^2 + y^2}$$

Magnitude $|z|$

$$|z| = r = \sqrt{x^2 + y^2}$$

Argument (θ)

$$\alpha = \tan^{-1}\left(\frac{y}{x}\right)$$



Quadrant 1:

$$\theta = \tan^{-1}\left(\frac{y}{x}\right)$$

$$\begin{matrix} x & +ve \\ y & +ve \end{matrix}$$

Quadrant 2:

$$\theta = \pi - \tan^{-1}\left(\frac{y}{x}\right)$$

$$\begin{matrix} x & -ve \\ y & +ve \end{matrix}$$

Quadrant 3:

$$\theta = -\pi + \tan^{-1}\left(\frac{y}{x}\right)$$

$$\begin{matrix} x & -ve \\ y & -ve \end{matrix}$$

Quadrant 4:

$$\theta = -\tan^{-1}\left(\frac{y}{x}\right)$$

$$\begin{matrix} x & +ve \\ y & -ve \end{matrix}$$

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$$\theta = \pi + \tan^{-1}\left(\frac{y}{x}\right)$$

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$$\theta = \pi + \tan^{-1}\left(\frac{y}{x}\right)$$

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$$\theta = 2\pi + \tan^{-1}\left(\frac{y}{x}\right)$$

Polar to Cartesian

$$r, \theta \rightarrow (x, y)$$

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$\therefore i = \sqrt{-1}$$

$$i^2 = -1$$

$$i^3 = -i$$

Complex Number in Polar form

$$z = a + bi \quad (\text{cartesian})$$

$$z = r e^{i\theta} \quad (\text{polar})$$

or

$$z = r (\cos \theta + i \sin \theta)$$

Miss PDF Questions

Q) $(-3, 5, 3)$ to cylindrical (r, θ, z)

$$r = \sqrt{x^2 + y^2}$$

$$\theta = \tan^{-1}\left(\frac{y}{x}\right)$$

$$r = \sqrt{(-3)^2 + (5)^2}$$

$$\theta = \pi + 1.01037$$

$$r = \sqrt{34}$$

$$\theta = 4.17$$

$$r = \sqrt{34}$$

$$= (\sqrt{34}, 4.17, 3)$$

Q) $(3, \frac{2\pi}{6}, -4)$ to cartesian (x, y, z)

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$x = 3 \cos\left(\frac{2\pi}{6}\right)$$

$$y = -\frac{3}{2}$$

$$x = -\frac{3\sqrt{3}}{2}$$

$$\left(-\frac{3\sqrt{3}}{2}, -\frac{3}{2}, -4\right)$$

Q) $(3, 4)$ to polar (r, θ)

$$r = \sqrt{x^2 + y^2}$$

$$\theta = \tan^{-1}\left(\frac{y}{x}\right)$$

$$r = \sqrt{3^2 + 4^2}$$

$$\theta = 0.927$$

$$r = \sqrt{25}$$

$$r = 5$$

Q) $(-5, 12)$ to polar (r, θ)

$$r = 13$$

$$\theta = \pi + \tan^{-1}\left(\frac{12}{-5}\right)$$

$$\theta = 1.7655$$

e) $(-7, -7)$

$$r = 7\sqrt{2}$$

$$\theta = \pi + \tan^{-1}\left(\frac{-7}{-7}\right)$$

$$\theta =$$

e) $(6, -8)$ 4th Quadrant

$$r = 10$$

$$\theta = 2\pi + \tan^{-1}\left(\frac{-8}{6}\right)$$

$$\theta = 5.355$$

e) Convert to polar:

1) $\left(5, \frac{\pi}{2}\right)$
r θ

$$x = r \cos \theta$$

$$y = \frac{5\sqrt{2}}{2}$$

$$x = \frac{5}{2}$$

$$\left(\frac{5}{2}, \frac{5\sqrt{2}}{2}\right)$$

2) $(5, 210) \longrightarrow \left(5, \frac{7\pi}{6}\right)$
r θ
 $= \left(\frac{-5\sqrt{3}}{2}, \frac{-5}{2}\right)$

Miss PDF argand diagrams

Q)

$$3 + 4i$$



Q) Convert into polar

i)

$$1 - \frac{\sqrt{3}}{4}i$$

$$r = \sqrt{1^2 + \left(\frac{\sqrt{3}}{4}\right)^2}$$

$$r = 2$$

$$\theta = 2\pi + \tan^{-1}\left(\frac{-\frac{\sqrt{3}}{4}}{1}\right)$$

$$\theta = \frac{5}{2}\pi$$

$$z = 2 \left(\cos\left(\frac{5}{2}\pi\right) + i \sin\left(\frac{5}{2}\pi\right) \right)$$

ii)

$$2$$

$$z = 2 + 0i$$

$$r = 2$$

$$\theta = \tan^{-1}\left(\frac{0}{2}\right)$$

$$\theta = 0$$

$$z = 2(1 + 0)$$

$$z = 2$$

iii) $-2 + 2i$

$$r = \sqrt{4+4}$$

$$\theta = \pi + \tan^{-1}\left(-\frac{2}{2}\right)$$

$$r = 2\sqrt{2}$$

$$\theta = \frac{3}{4}\pi$$

$$z = 2\sqrt{2} \left(\cos\left(\frac{3}{4}\pi\right) + i \sin\left(\frac{3}{4}\pi\right) \right)$$

iv) $-5i$

$$z = 0 - 5i$$

$$r = \sqrt{25}$$

$$\theta = 2\pi + \tan^{-1}\left(-\frac{5}{0}\right)$$

$$\therefore \tan^{-1}(\infty) = \frac{\pi}{2} \text{ or } 90^\circ$$

$$r = 5$$

$$\theta = 2\pi + \tan^{-1}(-\infty)$$

$$\tan^{-1}(-\infty) = -\frac{\pi}{2}$$

$$\theta = 2\pi + \left(-\frac{\pi}{2}\right)$$

$$\theta = \frac{3}{2}\pi$$

$$z = 5 \left(\cos\left(\frac{3}{2}\pi\right) + i \sin\left(\frac{3}{2}\pi\right) \right)$$

Q) Find z such that

$$|z| = 2 \quad \text{Arg } z = \frac{\pi}{4}$$

$$r = 2 \quad \theta = \frac{\pi}{4}$$

$$z = 2 \cos \frac{\pi}{4}$$

$$x = \sqrt{2} \quad y = \sqrt{2}$$

$$z = \sqrt{2} + \sqrt{2}i$$

Q) write down the modulus and argument of

i) $-\sqrt{3} + i$ $r = 2$ $\theta = \frac{5}{6}\pi$

ii) $4 + 4i$ $r = 4\sqrt{2}$ $\theta = \frac{\pi}{4}$

iii) Hence express the complex numbers $(-\sqrt{3} + i)(4 + 4i)$ and $\frac{-\sqrt{3} + i}{4 + 4i}$

$$\begin{aligned} \therefore r_1 r_2 (\cos \theta_1 + \cos \theta_2 + i \sin \theta_1 + i \sin \theta_2) \\ = r_1 r_2 (\cos (\theta_1 + \theta_2) + i \sin (\theta_1 + \theta_2)) \end{aligned}$$

$$\therefore \frac{r_1}{r_2} (\cos (\theta_1 - \theta_2) + i \sin (\theta_1 - \theta_2))$$

De Moivre's theorem

$$z^0 = 1$$

$$\therefore z = \cos \theta + i \sin \theta$$

$$z^{m+1} = z^m \cdot z$$

$$z^n = (\cos \theta + i \sin \theta)^n$$

$$z^{-m} = (z^{-1})^m$$

$$= \cos n\theta + i \sin n\theta$$

$$(z_1 z_2)^n = z_1^n \cdot z_2^n$$

o) $\left(\frac{\sqrt{3}-i}{\sqrt{3}+i} \right)^4$

first convert both to polar

$$\sqrt{3}-i, \quad r=2 \quad \theta = -\frac{\pi}{6}$$

$$\sqrt{3}+i, \quad r=2 \quad \theta = \frac{\pi}{6}$$

$$2 \left(\cos \left(-\frac{\pi}{6} \right) + i \sin \left(-\frac{\pi}{6} \right) \right), \quad 2 \left(\cos \left(\frac{\pi}{6} \right) + i \sin \left(\frac{\pi}{6} \right) \right)$$

$$= \frac{2^4 \left(\cos 4 \left(-\frac{\pi}{6} \right) + i \sin 4 \left(-\frac{\pi}{6} \right) \right)^4}{2^4 \left(\cos 4 \left(\frac{\pi}{6} \right) + i \sin 4 \left(\frac{\pi}{6} \right) \right)}$$

$$= \frac{-2^4}{-2^4}$$

$$= 1$$

e) $(1 - \sqrt{3}i)^{12}$

$$r = 2 \quad \theta = \frac{5}{3}\pi$$

$$z = 2 \left(\cos\left(\frac{5}{3}\pi\right) + i \sin\left(\frac{5}{3}\pi\right) \right)$$

$$= 2^{12} \left(\cos\left(12 \cdot \frac{5}{3}\pi\right) + i \sin\left(12 \cdot \frac{5}{3}\pi\right) \right)$$

$$= 2^{12} \left(\cos(20\pi) + i \sin(20\pi) \right)$$

$$= 4096$$

e) $\left(\frac{1+i}{1-i}\right)^3$?

$$z = 1 + i$$

$$r = \sqrt{2}$$

$$\theta = \frac{\pi}{4}$$

$$\sqrt{2} \left(\cos\left(\frac{\pi}{4}\right) + i \sin\left(\frac{\pi}{4}\right) \right)$$

$$z = 1 - i$$

$$r = \sqrt{2}$$

$$\theta = \frac{7}{4}\pi$$

$$\sqrt{2} \left(\cos\left(\frac{7}{4}\pi\right) + i \sin\left(\frac{7}{4}\pi\right) \right)$$

$$= \frac{\sqrt{2}^3 \left(\cos\left(\frac{3}{4}\pi\right) + i \sin\left(\frac{3}{4}\pi\right) \right)^3}{\sqrt{2}^3 \left(\cos\left(\frac{21}{4}\pi\right) + i \sin\left(\frac{21}{4}\pi\right) \right)^3}$$

Power

$$= \frac{-\frac{\sqrt{2}}{2} + i \frac{\sqrt{2}}{2}}{-\frac{\sqrt{2}}{2} - i \frac{\sqrt{2}}{2}}$$

$$= \frac{-\frac{\sqrt{2}}{2} + i \frac{\sqrt{2}}{2}}{-\frac{\sqrt{2}}{2} - i \frac{\sqrt{2}}{2}}$$

$$= \frac{-\frac{\sqrt{2}}{2} (1 - i)}{-\frac{\sqrt{2}}{2} (1 + i)}$$

$$= \frac{(1 - i)}{(1 + i)}$$

$$= \frac{1-i}{1+i} \cdot \frac{1-i}{1-i}$$

$$= \frac{1^2 - i^2}{1^2 + 2i + i^2}$$

$$= \frac{1 - (-1)}{1 + 2i - 1}$$

$$= \frac{2}{2i}$$

$$= \left(\frac{1}{i}\right)^3$$

$$= \frac{1}{i^3}$$

$$= \frac{1}{-i}$$

Application of De Moivre's Theorem

$$\cos(n\theta) + i \sin(n\theta) = (\underbrace{\cos\theta}_a + i \underbrace{\sin\theta}_b)^n$$

$$a = \cos\theta$$

$$b = i \sin\theta$$

Apply $(a+b)^n$

$$\underbrace{(\cos\theta + i \sin\theta)^n}_{\text{real}} = \underbrace{\cos^n\theta}_{\text{real}} + \underbrace{n C_1 \cos^{n-1}\theta (i \sin\theta)}_{\text{img}} + \underbrace{n C_2 \cos^{n-2}\theta (i \sin\theta)^2}_{\text{real}} + \underbrace{n C_3 \cos^{n-3}\theta (i \sin\theta)^3}_{\text{img}}$$

then simplify $i = i$, $i^2 = -1$, $i^3 = -i$

$$= \cos^n\theta + n C_1 \cos^{n-1}\theta (i \sin\theta) + n C_2 \cos^{n-2}\theta (-\sin^2\theta) + n C_3 \cos^{n-3}\theta (-i \sin^3\theta) + n C_4 \cos^{n-4}\theta (\sin^4\theta)$$

$$= [\cos^n\theta - n C_2 \cos^{n-2}\theta \sin^2\theta + n C_4 \cos^{n-4}\theta \sin^4\theta] + i [n C_1 \cos^{n-1}\theta \sin\theta - n C_3 \cos^{n-3}\theta \sin^3\theta]$$

separating the real and imaginary parts

$$\cos n\theta = [\cos^n\theta - n C_2 \cos^{n-2}\theta \sin^2\theta + n C_4 \cos^{n-4}\theta \sin^4\theta]$$

$$\sin n\theta = [n C_1 \cos^{n-1}\theta \sin\theta - n C_3 \cos^{n-3}\theta \sin^3\theta]$$

Q) $\cos 3\theta$

$$n = 3$$

$$= \cos^3\theta - 3 C_2 \cos\theta \sin^2\theta$$

$$= \cos^3\theta - 3 \cos\theta [1 - \cos^2\theta]$$

$$= \cos^3\theta - 3 \cos\theta + 3 \cos^3\theta$$

$$= 4 \cos^3\theta - 3 \cos\theta$$

Q) $\cos 4\theta$

$$\therefore \cos n\theta = [\cos^n\theta - n C_2 \cos^{n-2}\theta \sin^2\theta + n C_4 \cos^{n-4}\theta \sin^4\theta]$$

$$= \cos^4\theta - 4 C_2 \cos^2\theta \sin^2\theta + 4 C_4 \cos^0\theta \sin^4\theta$$

$$= \cos^4\theta - 4 \cos^2\theta (1 - \cos^2\theta) + (1 - \cos^2\theta)^2$$

$$= \cos^4\theta - 4 \cos^2\theta + 4 \cos^4\theta + \cos^4\theta - 2 \cos^2\theta + 1$$

$$(1 - \cos^2\theta)(1 - \cos^2\theta)$$

$$= 1 - \cos^2\theta - \cos^2\theta + \cos^4\theta$$

$$\therefore = \cos^4\theta - 2 \cos^2\theta + 1$$

$$= 8 \cos^4\theta - 8 \cos^2\theta + 1$$

Q) $\sin 4\theta$

$$\sin \theta = [{}^n C_1 \cos^{n-1} \theta \sin \theta - {}^n C_3 \cos^{n-3} \theta \sin^3 \theta]$$

Q) $\tan 6\theta$

$$= \frac{\sin 6\theta}{\cos 6\theta} \quad \therefore \sin \theta = [{}^n C_1 \cos^{n-1} \theta \sin \theta - {}^n C_3 \cos^{n-3} \theta \sin^3 \theta]$$

$$\therefore \cos \theta = [\cos^n \theta - {}^n C_2 \cos^{n-2} \theta \sin^2 \theta + {}^n C_4 \cos^{n-4} \theta \sin^4 \theta]$$

$$= \frac{{}^6 C_1 \cos^5 \theta \sin \theta - {}^6 C_3 \cos^3 \theta \sin^3 \theta + {}^6 C_5 \cos \theta \sin^5 \theta}{\cos^6 \theta - {}^6 C_2 \cos^4 \theta \sin^2 \theta + {}^6 C_4 \cos^2 \theta \sin^4 \theta - {}^6 C_6 \cos^0 \theta \sin^6 \theta}$$

$$= \frac{6 \cos^5 \theta \sin \theta - 20 \cos^3 \theta \sin^3 \theta + 6 \cos \theta \sin^5 \theta}{\cos^6 \theta - 15 \cos^4 \theta \sin^2 \theta + 15 \cos^2 \theta \sin^4 \theta - \sin^6 \theta}$$

$$= \frac{\frac{6 \cos^5 \theta \sin \theta}{\cos^6 \theta} - \frac{20 \cos^3 \theta \sin^3 \theta}{\cos^6 \theta} + \frac{6 \cos \theta \sin^5 \theta}{\cos^6 \theta}}{\frac{\cos^6 \theta}{\cos^6 \theta} - \frac{15 \cos^4 \theta \sin^2 \theta}{\cos^6 \theta} + \frac{15 \cos^2 \theta \sin^4 \theta}{\cos^6 \theta} - \frac{\sin^6 \theta}{\cos^6 \theta}}$$

$$= \frac{6 \tan \theta - 20 \tan^3 \theta + 6 \tan^5 \theta}{1 - 15 \tan^2 \theta + 15 \tan^4 \theta - \tan^6 \theta}$$

Q) $\tan 4\theta$

Roots of a Complex Number.

$$z_k = r^{1/n} \left[\cos \left(\frac{\theta + 2k\pi}{n} \right) + i \sin \left(\frac{\theta + 2k\pi}{n} \right) \right]$$

Q find the roots of $z^4 + 1 = 0$

$$z^4 + 1 = 0$$

$$z^4 = -1 + 0i \quad r = 1 \quad \theta = \pi$$

$$z^4 = (\cos \pi + i \sin \pi)$$

$$z = (\cos \pi + i \sin \pi)^{1/4}$$

(c) Express $\frac{(1-\sqrt{3}i)^6(2+2i)^3}{(1+\sqrt{3}i)^6}$ into form $a + ib$.

$$1 - \sqrt{3}i \quad r = 2, \quad \theta = \frac{5}{3}\pi$$

$$2(\cos(\frac{5}{3}\pi) + i \sin(\frac{5}{3}\pi))$$

$$2 + 2i \quad r = 2\sqrt{2}, \quad \theta = \frac{1}{4}\pi$$

$$2\sqrt{2}(\cos(\frac{1}{4}\pi) + i \sin(\frac{1}{4}\pi))$$

$$1 + \sqrt{3}i \quad r = 2, \quad \theta = \frac{1}{6}\pi$$

$$2(\cos(\frac{1}{6}\pi) + i \sin(\frac{1}{6}\pi))$$

$$= \frac{[2(\cos(\frac{5}{3}\pi) + i \sin(\frac{5}{3}\pi))]^6 [2\sqrt{2}(\cos(\frac{1}{4}\pi) + i \sin(\frac{1}{4}\pi))]^3}{[2(\cos(\frac{1}{6}\pi) + i \sin(\frac{1}{6}\pi))]^6}$$

$$= \frac{64(\cos(10\pi) + i \sin(10\pi)) \times 16\sqrt{2}(\cos(\frac{3}{4}\pi) + i \sin(\frac{3}{4}\pi))}{64(\cos(\frac{1}{2}\pi) + i \sin(\frac{1}{2}\pi))}$$

$$= \frac{64(1 + 0) \times 16\sqrt{2}(-\frac{\sqrt{2}}{2} + i \frac{\sqrt{2}}{2})}{64(0 + i)}$$

$$= \frac{64 \times 16\sqrt{2}(-\frac{\sqrt{2}}{2} + i \frac{\sqrt{2}}{2})}{64i}$$

$$= \frac{-16 + 16i}{-1}$$

$$= 16 - 16i$$